Practice problems #19

1. Find the determinant by row reduction to echelon form:

$$A = \begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ -2 & -6 & 2 & 3 & 9 \\ 3 & 7 & -3 & 8 & -7 \\ 3 & 5 & 5 & 2 & 7 \end{bmatrix}$$

2. Combine the methods of row reduction and cofactor expansion to compute the determinant:

$$A = \begin{bmatrix} -3 & -2 & 1 & -4 \\ 1 & 3 & 0 & -3 \\ -3 & 4 & -2 & 8 \\ 3 & -4 & 0 & 4 \end{bmatrix}$$

3. Let A and B be 4×4 matrices, with $\det A = -1$ and $\det B = 2$. Compute:

   (a) $\det AB$

   (b) $\det (B^5)$

   (c) $\det 2A$

4. True or False. Justify your answer.

   (a) Let A and B be square matrices, with B invertible. Then, $\det(BAB^{-1}) = \det A$.

   (b) Suppose that A is a square matrix such that $\det (A^4) = 0$. Then, columns of A are linearly independent.

   (c) Let U be a square matrix such that $U^T U = I$. Then $\det U = \pm 1$.

Answers or Hints on page 2.

#1: 24
#2: 0
#3: (a) -2, (b) 32, (c) -16
#4: (a) T, (b) F (c) T. Justify!